

COMPSCI 4NL3:

Final Report

Winter 2026
Due: April 7th

1 Overview

For the final project report, your team should now have an implementation of a solution for your proposed problem along with several comparable experiments from others. The items you should include in the report are detailed in the following section. Use the template I have uploaded to Avenue and amend it to create the final report. Note that there is some flexibility if, e.g. you have more to say about preprocessing and less about motivation. There are many ways to fill these pages to show you have put the necessary effort into the final project. You could, for instance, focus more on feature engineering, or trying different model architectures. You could try different data augmentation techniques, methods for regularization, ensembles, or embeddings. If you are not sure your intended work constitutes enough work for the final submission, please check with your TAs during the tutorial sessions or consult with us on Teams.

2 Model for Your Task

Now it's your turn to develop models for your task. You should implement one model per group member. What models can you choose? Any that we've discussed in task that are not simple baselines. You might decide to see how well you can do with a logistic regression model with heavy feature engineering. You might want to try fine-tuning a BERT model. The approach should be non-trivial and you should be able to provide some justification about why you think it will outperform models that other groups developed for your task.

You should implement these models and compare them to the work of others on your task. Was your approach better or worse than expected? As long as you can justify your approach and discuss it in your report, it is not critical that your method outperforms the others. It should outperform the baselines. You will need to reflect on the approaches and write about what you would suggest people do next, if they were to continue working on your task.

3 Report

Your complete project must now include the following items. The format of the report will be a PDF document which includes the following items.

1. (10%) **Introduction** Your report should have an introduction. You may largely copy the motivation and problem description from your previous reports. Write about previous work on your task or the most related task you can find. You should include at least 3 citations. References should be integrated as part of the discussion of the context of your work. They should not simply be listed as a set of relevant papers with no explanation.
2. (10%) **Dataset** You should describe the dataset properties and any preprocessing operations you performed. If you are annotating data yourself, describe the annotation procedure you developed and followed. Describe any changes you made to the dataset in a subsection if you have changed your dataset since the progress report. Some of you found that your dataset was too difficult or too easy. What did you have to change about your data? Did you have to augment the data or move to a completely new source?
3. (5%) **Features and Inputs** Describe your model inputs. What feature engineering or representation learning was performed? Why did you add these features? Why does it make sense to include them? Did you perform any feature selection or augmentation? One good way to vary your experiments is to try different kinds of features as inputs. How did you

vary the features used for your experiments? Were your inputs mostly word embeddings? Were these part of the model or static pretrained embeddings?

4. (20%) **Implementation** Describe the model implementation. You should have several models or versions of your model that you have run on your data. You should have a simple baseline to compare it to (likely majority vote), and may have other baselines from related work. Your model should outperform your simple baseline. For most of you, your model should also outperform at least one other trained model baseline. If it does not outperform this baseline and you expected it to, why doesn't it? You should be able to provide an explanation. This explanation can also be part of your error analysis section. You do not have to outperform all models from previous work, but you should have an approach you implement for comparison. What was your loss function? Describe the optimization technique. If you implemented a complex model with many parts, you should consider providing ablations. This means different experiments where you only include one feature at a time so you can tell what feature is contributing more to the performance that you see.
5. (10%) **Evaluation** Describe the evaluation strategy. What are your train/validation/test splits (size and label distributions)? What metrics are you using for evaluation?
6. (20%) **Comparison to Others** How did your approach compare to those of your classmates? You should be able to put a table together with your simple baselines, methods from other class members, and your own approaches. Clearly label which is which and describe your classmates approaches. Are the results what you expected? Why or why not?
7. (10%) **Error Analysis** Describe how you systematically examine the errors your model makes and provide supporting figures, stats, examples (e.g., confusion matrices, qualitative sample of test cases with high error margins, etc). What does your model appear to be good at? What does it seem to be bad at? How does the performance of your models differ? What patterns do you notice in the errors your model seems to make? What do you think you could do to specifically address those issues if you were to continue working on this model? What would you recommend to people in the future working on this problem?
8. (5%) **Team Contributions** Fill in the section about team contributions with a few sentences describing what each team member contributed. Please write at least one sentence per team member to describe what they contributed to and do not simply write "all members contributed equally".

4 Deliverables

You should submit your written final report and code to A2L. Zip these files (zip, tar, gz are OK). You will be assessed based on the completion of the above requirements. Your report should contain all the required items with sufficient detail. The report should follow the provided LaTeX template formatting with at least 4 pages of content not including graphs and tables. In other words, if you include a graph, try to make up for that space in the length of the document by writing onto the next page. We won't be very strict about this, we just want to avoid the case where, e.g. someone submits a 4 page report and one whole page has images and no text. You can include extra L^AT_EX packages, but you should not significantly change the formatting of the provided template.